Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Enuresis: What to Know and How to Treat?

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Under the Auspices of the IAP Action Plan 2022

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16

Introduction

The word "enuresis" has become synonymous with bedwetting. There can be nothing further from the truth and we need to erase this kind of thinking.

Definition

Enuresis is defined as involuntary passage of urine beyond the age at which bladder control is deemed to be achieved. Mainly can be divided as:

- ☑ *Daytime enuresis*: Often missed and the one that every pediatrician should be aware of to be able to diagnose and treat appropriately. Most children in this group get treated for the "nonexistent"' urinary tract infections (UTIs)!
- ☑ *Nocturnal enuresis***:** The enuresis occurs predominantly at night when the child is fast asleep (including wetting when asleep in the day).

Enuresis can be clinically considered as follows:

- ☑ Predominantly daytime (overactive bladder)
- ☑ Both daytime and nocturnal (posterior urethral valves, tubular problems, and neurogenic bladder)
- ☑ Predominantly nocturnal (primary nocturnal enuresis)

Different causes of enuresis and their management is given in **Table 1**.

TABLE 1: Conditions presenting as enuresis and treatment.

	Overactive bladder	***Posterior urethral valves***	***Juvenile nephronophthisis***
Symptoms	Frequency, urgency, dysuria, and daytime wetting	Can present as day-time wetting and wetting at night in older boys	Can present as daytime polyuria/wetting and nocturnal enuresis
Differential diagnosis	Urinary tract infections (UTI), local problems such as vulval redness	Mistaken for UTI, other enuresis causes	Overactive bladder, diabetes insipidus, diabetes mellitus, and chronic kidney failure
Clinical clues	Prevoid wetting, small volumes of urine, wetting by a few drops of urine, not waking to drink water at night, usually not wetting at night, and parents complain "waits till last minute to void"	☑ May be using abdominal pressure to void ☑ Urinary stream may not be good ☑ Pre- and postvoid wetting	☑ Nocturnal enuresis may start late ☑ Drinks large amounts of water ☑ Family history (autosomal recessive)
Diagnostic clues	Ultrasound shows normal kidneys and bladder with low prevoid volume and emptying to completion	☑ Ultrasound may show dilated ureters and renal pelvis/pelves but sometimes may not ☑ High postvoid residue (even after second void)	☑ *Ultrasound:* Normal kidneys and bladder ☑ Urine shows low-specific gravity (1.005), may be trace sugar, minor hematuria and proteinuria
Investigation	Ultrasound scan of kidneys and bladder	Micturating cystogram (MCU)	Raised serum creatinine, hyposthenuria, raised urinary tubular protein, and characteristic kidney biopsy
Maneuvers	Vincent's curtsy, crossing legs, and holding perineum with hand or heel		
Aggravating factors	Constipation, stimulant drinks such as tea and coffee, and fizzy drinks		
Treatment	☑ Treat constipation (lactulose and lactitol) used for prolonged period (3–4 months) ☑ Stop stimulant and fizzy drinks ☑ Voiding diary for noting improvement ☑ Anticholinergic medication (oxybutynin 0.5–0.6 mg/kg/day in two or three doses, tolterodine 2 mg for a child <35 kg once a day)	Fulguration of posterior urethral valves (surgery) and long-term follow-up of renal function	Need dialysis and transplantation when end-stage kidney disease

TABLE 2: Conditions presenting as enuresis and treatment (*Continued*).

	Neurogenic bladder	***Ectopic ureter insertion below the bladder sphincter***	***Diabetes mellitus (D)/diabetes insipidus (DI)***	***Primary nocturnal enuresis***
Symptoms	☑ Wetting throughout ☑ Dribbling of urine ☑ Recurrent UTIs ☑ Severe constipation	☑ Continuous urinary leakage and wetting ☑ Never dry when awake or asleep	☑ Increased thirst ☑ Frequent voiding nocturnal enuresis weight loss, irritability, and behavior changes	☑ Wetting at night persistent above 5 years of age when asleep ☑ No daytime symptoms
Differential diagnosis	☑ Enuresis ☑ Posterior urethral valves ☑ Other causes of recurrent UTIs	☑ Enuresis ☑ Bladder dysfunction	Tubular disorders UTI, and chronic kidney disease	May be confused with overactive bladder, diabetes mellitus, and diabetes Insipidus
Clinical clues	☑ History of repair of meningomyelocele/spinal tumor/trauma ☑ Any lump or tuft of hair in the back ☑ Delayed walking/ abnormal gait ☑ Lower limb weakness ☑ Severe constipation	☑ Never dry ☑ Continuous leaking of urine often seen on vulval examination	☑ *DM:* Enuresis, weight loss despite polyphagia, and family history ☑ *DI:* Polydipsia and polyuria, waking at night for water and voiding urine ☑ *Central DI:* CNS insult history	Wetting at least 5 nights per week, no daytime symptoms, positive family history in parents, no polyuria or failure to thrive
Diagnostic clues	☑ *USG KUB:* Thick trabeculated bladder wall with often bilateral hydronephrosis ☑ *MCUG:* Thick walled trabeculated bladder and bilateral VUR and poor emptying ☑ MRI for spinal defect ☑ Associated urine infection	☑ Ultrasound shows normal kidneys and bladder ☑ Hydroureter may be seen ☑ Vesicoureteral reflux may be present	☑ *DM:* Glycosuria, high fasting/ random blood sugars, and raised Hb1Ac ☑ *DI:* Urine osmolality <300 mosm/L, hypernatremia if inability to drink	Normal USG KUB, normal urine examination
Investigation	☑ USG KUB ☑ MCUG ☑ Urodynamic study ☑ MRI lumbosacral spine	☑ USG KUB ☑ MR urography or Contrast CT KUB ☑ Cystoscopy	☑ *DM:* Blood sugar and Hb1Ac ☑ *DI:* Paired serum and urine osmolarity, water deprivation test	Proper history is diagnostic, investigations not required in majority
Treatment	☑ Anticholinergic treatment (oxybutynin, tolterodine as described above) ☑ Treat constipation ☑ Clean intermittent catheterization (CIC) ☑ Combined nephro-urology long-term management	*Surgical therapy:* Ureteric reimplantation	☑ *DM:* Dietary restrictions and insulin therapy ☑ *Central DI:* Vasopressin—intranasal/oral ☑ *Nephrogenic DI:* Reduce solute intake (salt/ proteins) to reduce obligatory urine water losses ☑ Thiazides/ amiloride/ indomethacin	☑ Optimization of fluids ☑ Avoid bladder irritants past evening ☑ Avoid constipation ☑ Star charts and night bed alarms ☑ *Primary Therapy:* ☑ Oral desmopressin for 3 months and reassess (>50% reduction in wetting episodes considered success) ☑ Enuresis alarm ☑ Combination therapy (if either monotherapy fails)

(CNS: central nervous system; Hb1Ac: hemoglobin A1c; MCUG: micturating cystourethrogram; USG KUB: ultrasonography for kidney, ureter, and urinary bladder; UTI: urinary tract infection)

Conclusion

There is varied etiology for enuresis. After identifying whether daytime or nocturnal, treatment should be directed at the cause of the enuresis.

Treatment (In Brief)

TABLE 3: Treatment of different conditions of enuresis.

Conditions	***Treatment (nonpharmacologic)***	***Treatment (pharmacologic)***
Detrusor instability	☑ Avoid constipation ☑ Avoid caffeinated drink and carbonated drinks ☑ Ensure no vulval redness in girls	☑ Lactulose/lactitol for prolonged periods (3–4 months) ☑ Tablet oxybutynin (0.5–0.6 mg/kg/day in two to three doses or tablet tolterodine 2 mg (weight < 35 kg)
PU valves	Fulguration of valves	
Juvenile nephronophthisis	Management of chronic kidney disease	
Neurogenic bladder	☑ Avoid constipation ☑ Clean intermittent catheterization (CIC)	Lactulose/lactitol as above Oxybutynin/tolterodine as above
Ectopic ureter	Surgery	
Diabetes mellitus/ insipidus	Specific therapy like insulin/ antidiuretic hormone (ADH)	*Central diabetes insipidus (DI):* ☑ Oral desmopressin—0.05–0.2 mg/day BD/TDS ☑ Intranasal 5–30 µg/day BD/TDS *Nephrogenic DI:* ☑ Thiazides—2–4 mg/kg/day in three divided doses ☑ Amiloride—used along with thiazides, 0.3 mg/kg/ day in three divided doses ☑ Indomethacin (use with caution, risk of renal injury) —2–3 mg/kg day in two to three divided doses
Primary nocturnal enuresis	☑ Treat constipation ☑ Ensure less fluid intake after 7 PM ☑ Ensure adequate fluid intake in the day ☑ Avoid caffeinated drinks and fizzy drinks	*Oral desmopressin:* 0.2–0.4 mg—1 hour before sleep for months—taper and stop

Further Reading

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Sinusitis

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